

Homework 1

For this homework you will create a github repo, set up github pages, clone the repo to your computer as an R **project**, create a `.qmd` file, and push those changes back to github to create a webpage! You'll submit the link to your github pages site (the one that renders as a nice website).

Sounds like a lot. We'll get through it! (We're going to follow the process outlined for 'render to docs' [from here](#).)

Step 1

- Head to github and create a new repo.
 - Be sure to make the repo public and **do not** choose a `.gitignore` for this one

Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository](#).

Required fields are marked with an asterisk (*).

Repository template

No template ▾

Start your repository with a template repository's contents.

Owner *

 jbpost2 ▾

Repository name *

Homework1

✔ Homework1 is available.

Great repository names are short and memorable. Need inspiration? How about [refactored-dollop](#) ?

Description (optional)

☒



Public

Anyone on the internet can see this repository. You choose who can commit.

☐



Private

You choose who can see and commit to this repository.

Initialize this repository with:

☒ Add a README file

This is where you can write a long description for your project. [Learn more about READMEs](#).

Add .gitignore

.gitignore template: None ▾

Choose which files not to track from a list of templates. [Learn more about ignoring files](#).

Choose a license

License: Apache License 2.0 ▾

A license tells others what they can and can't do with your code. [Learn more about licenses](#).

This will set  `main` as the default branch. Change the default name in your [settings](#).

① You are creating a public repository in your personal account.

Figure 1: Github repo creation form.

Step 2

- Create a new **R project** from version control (as we did in the notes/videos) that clones this repository locally.
 - Recall there is a (green) ‘<> Code’ button to copy the repo link.
 - A `.gitignore` file may be created in this process. That isn’t a worry!

Step 3

- Within **RStudio**, now create a new `.qmd` document that outputs to **HTML**. Give this a title about data science. Save the file in the main repo folder on your computer.
- In this **quarto** document, we want to do two things (separate them with headers):
 - First, write text to answer the question:
 - * What do you think being a data scientist is about?
 - * What differences/similarities do you see between data scientists and statisticians?
 - * How do you view yourself in relation to these two areas?
 - You can write in a conversational tone or more formally (however you want to represent yourself). **Use a markdown list at some point.** There is no word count or anything like that, just make sure you answer the prompts above to receive full credit.
 - Second, include a section with the following R code:

```
y <- density(iris$Petal.Width)
```

- * Create an R code chunk to determine the class, type, and structure of the object `y`
 - * Create an R code chunk that uses the `plot` function on `y`. Hide the R code in the final document by setting `echo` to `FALSE`.
 - * Include some markdown text between these code chunks explaining what you are doing with the R code.
- You can render the document as needed to check things are looking good locally.

Step 4

In your repo folder on your computer, create a file called `_quarto.yml`. Open this file (perhaps in **RStudio** or a text editor) and place the following in the file (spacing is important - the spacing must match below exactly - two spaces prior to `type` and `output-dir`!):

```
project:
  type: website
  output-dir: docs
```

To create the file, you can navigate to the folder and create a new file.

- On windows, first make sure that file extensions show when you look at files in folders.
 - Now right click in the folder area, select ‘New’ -> ‘Text Document’. Change the file name and file extension (the `.txt` part) to `_quarto.yml`. Now you can open the file in **RStudio** or a text editor and add the text.
- On a mac, you should be able to create a file using **TextEdit** in your **R project** folder. Save it as a `.doc` or whatever. Then you should be able to rename the file as `_quarto.yml`. Now you can open the file in **RStudio** or a text editor and add the text.

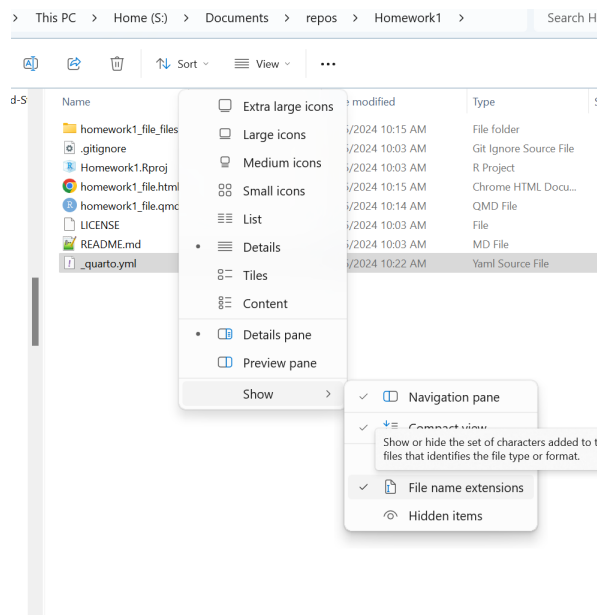


Figure 2: Show File Name Extensions

Step 5

Now you can use a similar process to create a file called `.nojekyll` in the project repo on your computer. This file doesn't need to have anything in it! You just need that file there. Note: On Windows, once you make this file, it may not show (files that start with a period are 'hidden' files). Don't worry, `git` still tracks this file. You can also go to the same 'View' menu as before and select 'Hidden items' to show these types of files.

Step 6

Now head back to RStudio. Open the terminal in RStudio (near the console there should be a few tabs, choose the Terminal). In that terminal, run the following code:

```
quarto render
```

Step 7

We did everything we need. We've set up the `.nojekyll` file to keep GitHub from render the main repo and we've created the `_quarto.yml` file to output our quarto files in a `doc` folder. Now we want to push all changes up to your repo on GitHub! You can do this via menus in RStudio or via the command line. If you can't get either of those to work, you can use the GitHub web interface for now.

Step 8

Now we just need to change a few things on GitHub. Head to your GitHub repo page. Go to settings, choose pages, and under "Branch" choose 'main' and change the folder to '/docs'. Then hit save!

```
Console Terminal Background Jobs
Terminal 1 MINGW64/s/Documents/repos/Homework1
jbpst2@ST-5272PC01 MINGW64 /s/
Documents/repos/Homework1 (main)
$ quarto render

processing file: homework1_file
.qmd
|
|
|.....
|.....| 20% |.....| 40% |.....|
| 60% |.....| 80% |.....| 100% |
|.....|.....|.....|.....|.....|

output file: homework1_file.knit.md

pandoc
to: html
output-file: homework1_file.html
standalone: true
section-divs: true
html-math-method: mathjax
wrap: none
default-image-extension: png

metadata
document-css: false
link-citations: true
date-format: long
lang: en
title: Data Science Thoughts
editor: visual

Output created: docs\homework1_file.html

jbpst2@ST-5272PC01 MINGW64 /s/Documents/repos/Homework1 (main)
$
```

Figure 3: Image of the RStudio terminal after running quarto render

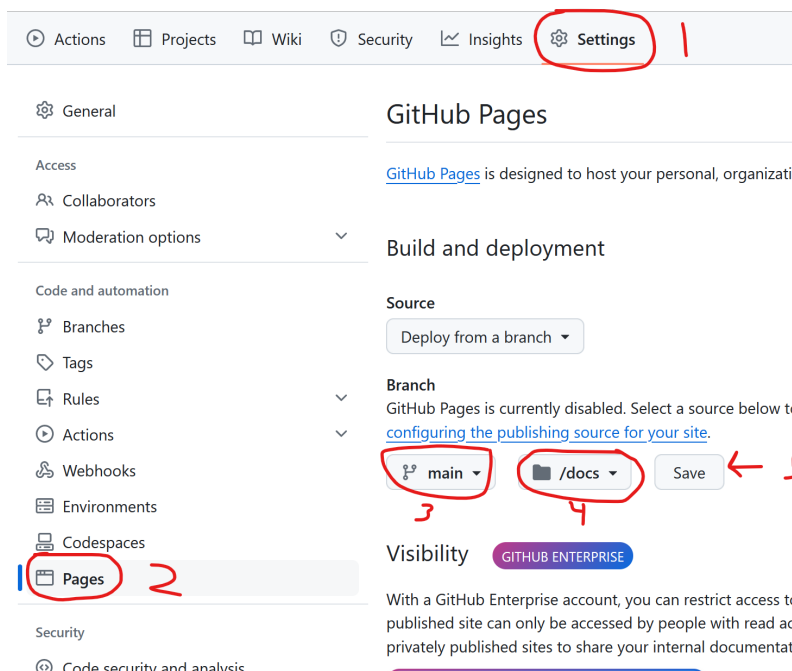


Figure 4: Options to choose in the Settings of your GitHub repo.

Step 9

Here we have to be patient! Wait about 2 minutes then head back to your main GitHub repo page. There will now be a ‘Deployments’ section on the bottom right of the page.

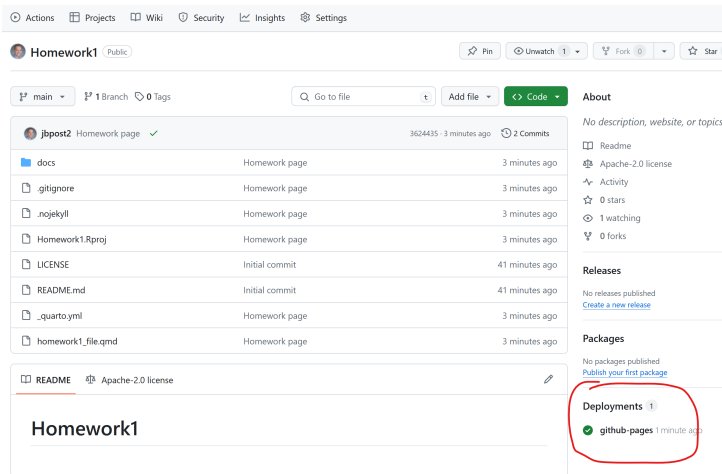


Figure 5: Deployment section shown.

Click on that. Deployments of your ‘rendered’ website are listed here. A green check mark appears next to the deployment when it is ready! The link below that is the link to the ‘rendered’ site.

github-pages deployments

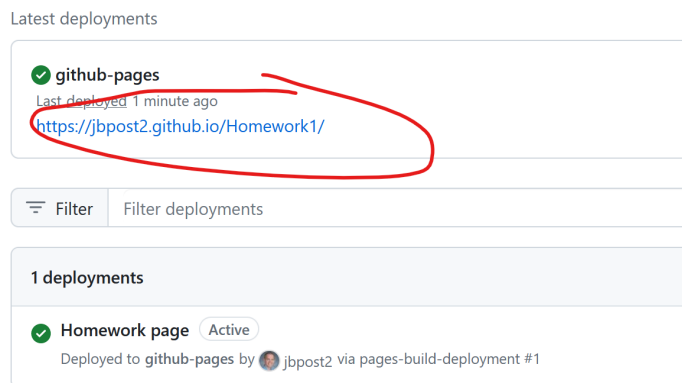


Figure 6: Link to website.

Copy the link to that site and turn this in!